6.1 Numbers to 100,000: answers

← Unit overview · Practice

For parents: when an answer differs, ask “How did you get that?” before showing this sheet. Often the method is fine and there’s one small slip.

Set 1 · Warm-up

#	Answer	How
1	50	The 5 is in the tens column: 5 tens = 50.
2	462	H T E = 4 6 2.
3	2 400	2 thousands, 4 hundreds, 0 tens, 0 ones.
4	1 325	One digit in each column.
5	4 000	All the ones, tens and hundreds are “full”, so the thousands go up by one.
6	4 999	One less than 5 000.
7	2 460	Only the tens digit changes.
8	6 820	Only the hundreds digit changes.
9	4 098	Same thousands and hundreds; 9 tens is more than 8 tens.
10	7 000 + 300 + 80 + 4	D H T E.

Set 2 · Warm-up → Building

#	Answer	How
1	40 000	The 4 is in the ten-thousands (TD) column.
2	300	The 3 is in the hundreds column.
3	26 500	26 thousands and 5 hundreds.
4	70 080	7 ten-thousands, 0 thousands, 0 hundreds, 8 tens, 0 ones: zeros fill the empty columns.
5	eighteen thousand four hundred (*achttienduizend vierhonderd*)	18 thousands and 4 hundreds.
6	34 607	Each part goes in its own column; 0 tens.
7	50 209	0 thousands and 0 tens need zeros.

#	Answer	How
8	24 500	The thousands digit goes from 3 to 4.
9	77 890	The ten-thousands digit goes from 6 to 7.
10	39 200	40 thousands minus 1 thousand is 39 thousands.
11	82 051	Read the table from left to right.

Set 3 · Building: the number line

#	Answer	How
1	5 000	Half of 10 000.
2	25 000	Halfway between 20 and 30 is 25, so 25 thousand.
3	3 450	Halfway between 400 and 500 is 450.
4a	100	1 000 is split into 10 equal steps: 1 000 ÷ 10 = 100.
4b	A = 6 100, B = 6 500, C = 6 800	Count the steps: A is 1 step, B is 5 steps (halfway), C is 8 steps.
5	48 000, 49 000	Counting on in thousands.
6	12 100, 12 000	Counting back in hundreds.
7	35 000, 45 000	Counting on in ten-thousands; only the first digit changes.
8	70 000	69 990 is 10 below 70 000 and 70 010 is 10 above.
9	25 a quarter of the way along, 50 in the middle, 90 close to 100	Any sensible placement is fine. Ask how he decided.

Set 4 · Building: rounding

#	Answer	How
1	50	47 is between 40 and 50, closer to 50 (7 ones → round up).
2	380	2 ones → round down.
3	400	Look at the tens: 8 tens → round up.
4	1 300	1 250 is exactly halfway; the rule is that halfway rounds up.

#	Answer	How
5	6 000	Look at the hundreds: 4 → round down.
6	7 000	5 hundreds → round up.
7	24 000	7 hundreds → round up.
8	20 000	Look at the thousands: 3 → round down.
9	To the nearest 1 000	To the nearest 100 it would be 18 700; to the nearest 1 000 it's 19 000.
10	€500	8 tens → round up.

Set 5 · Stretch

#	Answer	How
1	4 050 < 40 050 < 40 500 < 45 000	4 050 has only four digits. The others: compare thousands (0, 0, 5), then hundreds (0 vs 5).
2a	75 310	Biggest digit first.
2b	10 357	Smallest digit first, but a five-digit number can't start with 0, so put 1 first and the 0 second.
3	Sunday, 90 more	12 540 − 12 450 = 90. (Count up from 12 450 to 12 500 is 50, then 40 more.)
4	64 250	Thousands digit goes from 3 to 4.
5	53 020	40 000 + 13 000 = 53 000, plus 20.
6	45	4 500 = 45 × 100. Think €4 500 in €100 notes.
7	30	30 000 = 30 × 1 000.
8a	4 500	Halfway rounds up, so 4 500 rounds to 5 000. 4 499 would round to 4 000.
8b	5 499	5 500 would round up to 6 000.
9	20 000	9 500 + 500 = 10 000, plus 10 000 more.
10	100 000	Every column is full, so they all become 0 and a new column starts: one hundred thousand.

Puzzle corner

24 numbers. To round to 30 000, a number must be from 25 000 up to 34 999.

- Starting with **2**: the next digit must be 5 (25…), then 1, 3, 4 in any order: 6 numbers (25 134, 25 143, 25 314, 25 341, 25 413, 25 431).
- Starting with **3**: the next digit can be 1, 2 or 4, but not 5 (35… is too big and rounds to 40 000). For each, the other three digits can go in 6 orders: $3 \times 6 = 18$ numbers.
- Total $6 + 18 = 24$.

Any correct number he finds is a success. Listing them systematically (in order) is the “big challenge” skill, so praise an organised list even if it isn’t complete.